Syllabus Year: 2023-2024

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - IV

Course Code: BCA431 **Max Marks: 100**

Course Name: GRAPH THEORY **Duration: 03:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Enhance research, inquiry and analytical thinking abilities.

CO2: To improve proof-writing skills.

CO3: Apply the basics of Graph Theory to solving practical problems.

Section A **3*10=30**

1)	Discuss how graph theory can be applied to scheduling final examinations in a university setting.	**3**	(CO3, L2)
2)	Find the number of edges in a simple graph with 8 vertices where each vertex has a degree of 3.	**3**	(CO1, L3)
3) a)	Explain how the number of edges in a simple graph can be calculated using the degree sum formula and illustrate your explanation with an example.	**3**	(CO1, L2)
	OR		
b)	Describe how graph theory can be applied to determine the optimal encoding scheme for symbols based on their frequencies.	**3**	(CO1, L2)
4)	Explain how a graph can be used to determine if there exists an Eulerian path or circuit. Illustrate your explanation with an appropriate example.	**3**	(CO3, L2)
5)	Analyze the difference between a Hamiltonian cycle and an Eulerian circuit. Provide an example of each.	**3**	(CO1, L4)
6)	Analyze how graph theory can be applied to determine the minimum spanning tree in a network of 10 cities connected by roads with varying distances between them. What is the significance of such an application?	**3**	(CO1, L4)
7)	Determine whether a given graph is a tree or not by checking for the necessary conditions. Demonstrate your answer with an example graph.	**3**	(CO2, L3)
8) a)	Determine whether a given graph can have more than one minimum spanning tree, and provide an example if possible.	**3**	(CO2, L3)
	OR		
b)	Demonstrate how to use graph theory to determine if a given scheduling problem can be solved using a bipartite graph.	**3**	(CO2, L3)

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9)	Construct a binary tree for the given symbols with their respective frequencies and calculate the average number of bits required to encode these symbols: : 0.15, : 0.35, : 0.20, : 0.20.	**3**	(CO1, L3)
10)	Analyze how graph theory can be applied to determine the minimum number of test cases required for software testing using graph coloring.	**3**	(CO2, L4)

Section B **7*3=21**

11)	Analyze a simple graph with 8 vertices and determine whether it can have exactly 15 edges without containing a subgraph that is a complete graph on 4 vertices (K_4). Justify your answer using graph theory principles.	**7**	(CO2, L4)
12) a)	Discuss the concept of a spanning tree in graph theory and its applications. (i) Explain what is meant by a spanning tree of a connected graph. (ii) Describe an algorithm to find a minimum spanning tree for a given weighted simple graph with 8 vertices and 12 edges, and illustrate it using a step-by-step approach. (iii) Discuss the importance of spanning trees in network design and provide an example of their practical application.	**7**	(CO3, L2)
	OR		
b)	Discuss how graph theory can be applied to optimize delivery routes for a logistics company by addressing (i) finding an optimal route using Dijkstra's algorithm, (ii) determining the minimum spanning tree to minimize transportation costs, and (iii) identifying critical nodes that could cause delays.	**7**	(CO3, L2)
13)	Discuss how to determine if a given graph is a tree or not. Additionally, provide an algorithm to find a spanning tree for a connected, undirected graph G with n vertices. Prove that your algorithm always returns a minimum spanning tree if all edge weights in G are distinct.	**7**	(CO2, L2)

Section C **7*4=28**

14)	Construct a graph with 8 vertices that does not contain any cycles of length 4 (i.e., no squares). Determine the maximum number of edges such a graph can have and justify your answer. Additionally, explain how this graph can be used to solve a practical problem related to network design.	**7**	(CO3, L3)
15)	Demonstrate how to construct a spanning tree for the given graph G using Prim's algorithm. Then, show whether the constructed spanning tree is unique by considering a different starting vertex. The graph G has 7 vertices labeled A, B, C, D, E, F, and G with the following edge weights: AB (2), AC (3), AD (4), BC (1), BD (5), CD (6), CE (2), DE (3), EF (4), FG (3), EG (5).	**7**	(CO2, L3)
16) a)	Construct a graph representing a scheduling problem for final exams in a university course, ensuring that no two courses with overlapping times are scheduled at the same time slot. Using your constructed graph, (i) determine if there exists a valid schedule by finding a maximum matching, (ii) color the vertices of this graph using the minimum number of colors such that no two adjacent vertices share the same color, and (iii) justify whether the exam schedule can be optimized further based on the chromatic index. Demonstrate your solution with appropriate diagrams or tables for clarity.	**7**	(CO3, L3)
	OR		

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b)	A network engineer is tasked with designing a secure and efficient communication network for a new office building that has 10 rooms (vertices). The engineer wants to ensure that every room can communicate directly or indirectly through other rooms, but no cycles should exist in the network to avoid potential security breaches. Using graph theory principles, determine whether the following design meets these requirements: Construct a graph with 10 vertices where each vertex represents a room and an edge between two vertices indicates direct communication. The edges are given as follows: (1-2), (1-3), (2-4), (2-5), (3-6), (3-7), (4-8), (4-9), (5-10). Demonstrate your answer by identifying if the graph is a tree and finding its spanning tree, if it exists. If the graph is not a tree, suggest a minimum number of edges to add or remove to make it a tree while ensuring all rooms are connected.	**7**	(CO3, L3)
17)	Analyze a given graph to determine whether it can have more than one distinct spanning tree and explain your findings. Specifically, consider the following graphs: Graph G1 with vertices {A, B, C, D, E} and edges {(A,B), (B,C), (C,D), (D,E), (E,A)}, and Graph G2 with vertices {P, Q, R, S, T} and edges {(P,Q), (Q,R), (R,S), (S,T), (T,P), (P,R)}. For each graph, provide a detailed explanation of the spanning trees present, and justify your answer using relevant concepts from tree theory.	**7**	(CO1, L4)

Section D **7*3=21**

18)	Determine whether a given graph with 9 vertices and 12 edges can be both Eulerian and Hamiltonian. If not possible, identify which property (Eulerian or Hamiltonian) it cannot satisfy and provide a reason for your answer. Construct an example of such a graph if it is feasible.	**7**	(CO3, L3)
19) a)	Discuss how graph theory can be applied to optimize network routing in a university campus network. In your answer, explain (i) the concept of a minimum spanning tree and its relevance to network design; (ii) describe Dijkstra's algorithm for finding the shortest path between two nodes in a weighted graph; (iii) illustrate these concepts using an example of a small-scale campus network with at least 5 buildings.	**7**	(CO3, L2)

OR

b)	Illustrate how graph theory can be applied to optimize the frequency assignment in a mobile network. Specifically, consider a network with 10 base stations where each station has a unique set of frequencies assigned based on its traffic load. Describe a method using graph coloring to ensure that no two adjacent (i.e., overlapping coverage areas) base stations are assigned the same frequency. Calculate the minimum number of colors needed for this assignment and explain why your choice is optimal. Additionally, discuss how you would determine if an alternate set of 10 frequencies can be used more efficiently.	**7**	(CO3, L2)
20)	Analyze an 8-vertex tree and determine whether it is a full ary tree for each possible value of ary (ary from 2 to 4). For each case, justify your answer by calculating the minimum number of vertices required for such a structure and comparing it with the given number of vertices. Also, compare the properties of these trees in terms of their height and number of non-pendant vertices.	**7**	(CO1, L4)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember

L2 - Understand

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L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create